A Practical Guide to Masked language model

TOPIC -- MASKED LANGUAGE MODEL

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Tokenizer: This module converts a piece of English text into a sequence of integers ("tokens"). Embedding: This module converts the sequence of tokens into an array of real-valued vectors representing the tokens. It represents the conversion of discrete token types into a lower-dimensional Euclidean space. Encoder: a stack of Transformer blocks with self-attention, but without causal masking. Task head: This module converts the final representation vectors into one-shot encoded tokens again by producing a predicted probability distribution over the token types. It can be viewed as a simple decoder, decoding the latent representation into token types, or as an "un-embedding layer". The task head is necessary for pre-training, but it is often unnecessary for so-called "downstream tasks," such as question answering or sentiment classification. Instead, one removes the task head and replaces it with a newly initialized module suited for the task, and finetune the new module. The latent vector representation of the model is directly fed into this new module, allowing for sample-efficient transfer learning.

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the feed-forward size and filter size are synonymous. Both of them denote the number of dimensions in the middle layer of the feed-forward network. the hidden size and embedding size are synonymous. Both of them denote the number of real numbers used to represent a token. The notation for encoder stack is written as L/H. For example, BERTBASE is written as 12L/768H, BERTLARGE as 24L/1024H, and BERTTINY as 2L/128H.

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Given "[CLS] my dog is cute [SEP] he likes playing [SEP]", the model should predict [IsNext]. Given "[CLS] my dog is cute [SEP] how do magnets work [SEP]", the model should predict [NotNext].

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Embedding This section describes the embedding used by BERTBASE. The other one, BERTLARGE, is similar, just larger. The tokenizer of BERT is WordPiece, which is a sub-word strategy like byte-pair encoding. Its vocabulary size is 30,000, and any token not appearing in its vocabulary is replaced by

[UNK] ("unknown").

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Bidirectional encoder representations from transformers (BERT) is a language model introduced in October 2018 by researchers at Google. It learns to represent text as a sequence of vectors using self-supervised learning. It uses the encoder-only transformer architecture. BERT dramatically improved the state of the art for large language models. As of 2020, BERT is a ubiquitous baseline in natural language processing (NLP) experiments. BERT is trained by masked token prediction and next sentence prediction. With this training, BERT learns contextual, latent representations of tokens in their context, similar to ELMo and GPT-2. It found applications for many natural language processing tasks, such as coreference resolution and polysemy resolution. It improved on ELMo and spawned the study of "BERTology", which attempts to interpret what is learned by BERT. BERT was originally implemented in the English language at two model sizes, BERTBASE (110 million parameters) and BERTLARGE (340 million parameters). Both were trained on the Toronto BookCorpus (800M words) and English Wikipedia (2,500M words). The weights were released on GitHub. On March 11, 2020, 24 smaller models were released, the smallest being BERTTINY with just 4 million parameters.

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In masked language modeling, 15% of tokens would be randomly selected for masked-prediction task, and the training objective was to predict the masked token given its context. In more detail, the selected token is:

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Architectural family The encoder stack of BERT has 2 free parameters: L {\displaystyle L} , the number of layers, and H {\displaystyle H} , the hidden size. There are always H / 64 {\displaystyle H/64} self-attention heads, and the feed-forward/filter size is always 4 H {\displaystyle 4H} . By varying these two numbers, one obtains an entire family of BERT models. For BERT:

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Variants The BERT models were influential and inspired many variants. RoBERTa (2019) was an engineering improvement. It preserves BERT's architecture (slightly larger, at 355M parameters), but improves its training, changing key hyperparameters, removing the next-sentence prediction task, and using much larger mini-batch sizes. XLM-RoBERTa (2019) was a multilingual RoBERTa model. It was one of the first works on multilingual language modeling

at scale. DistilBERT (2019) distills BERTBASE to a model with just 60% of its parameters (66M), while preserving 95% of its benchmark scores. Similarly, TinyBERT (2019) is a distilled model with just 28% of its parameters. ALBERT (2019) used shared-parameter across layers, and experimented with independently varying the hidden size and the word-embedding layer's output size as two hyperparameters. They also replaced the next sentence prediction task with the sentence-order prediction (SOP) task, where the model must distinguish the correct order of two consecutive text segments from their reversed order. ELECTRA (2020) applied the idea of generative adversarial networks to the MLM task. Instead of masking out tokens, a small language model generates random plausible substitutions, and a larger network identify these replaced tokens. The small model aims to fool the large model. DeBERTa (2020) is a significant architectural variant, with disentangled attention. Its key idea is to treat the positional and token encodings separately throughout the attention mechanism. Instead of combining the positional encoding (x p o s i t i o n {\displaystyle x_{\mathrm {position} }}) and token encoding (x t o k e n {\displaystyle x_{\mathrm {token} }}) into a single input vector (x i n p u t = x p o s i t i o n + x t o k e n {\displaystyle x_{\mathrm {input} }=x_{\mathrm {position} }+x_{\mathrm {token} }}), DeBERTa keeps them separate as a tuple: (x p o s i t i o n , x t o k e n) {\displaystyle (x_{\mathrm {position} },x_{\mathrm {token} })} . Then, at each self-attention layer, DeBERTa computes three distinct attention matrices, rather than the single attention matrix used in BERT:

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Masked language modeling (MLM): In this task, BERT ingests a sequence of words, where one word may be randomly changed ("masked"), and BERT tries to predict the original words that had been changed. For example, in the sentence "The cat sat on the [MASK]," BERT would need to predict "mat." This helps BERT learn bidirectional context, meaning it understands the relationships between words not just from left to right or right to left but from both directions at the same time. Next sentence prediction (NSP): In this task, BERT is trained to predict whether one sentence logically follows another. For example, given two sentences, "The cat sat on the mat" and "It was a sunny day", BERT has to decide if the second sentence is a valid continuation of the first one. This helps BERT understand relationships between sentences, which is important for tasks like question answering or document classification.